Set-A Question

(a) inverting Schmitt Trigger

(b) Higher threshold, $V_{TH} = \frac{R_1}{R_1+R_2} V_H$

$= \frac{20k}{20k+30k} \times 10V$

$\boxed{V_{TH} = 4V}$

Lower threshold, $V_{TL} = \frac{R_1}{R_1+R_2} V_L$

$= \frac{20k}{20k+30k} \times -5V$

$\boxed{V_{TL} = -2V}$

(c) High time $T_1 = R_x C_x \ln\left|\frac{V_H - V_{TL}}{V_H - V_{TH}}\right|$

$= R_x C_x \ln\left(\frac{10+2}{10-4}\right)$

$= R_x C_x \ln 2$

Low time, $T_2 = R_x C_x \ln\left|\frac{V_L - V_{TH}}{V_L - V_{TL}}\right|$

$= R_x C_x \ln\left(\frac{-5-4}{-5+2}\right)$

$= R_x C_x \ln 3$

Duty cycle, $D = \frac{T_1}{T_1+T_2} = \frac{\ln 2}{\ln 2 + \ln 3}$

$= 0.3868 = \boxed{38.69\%}$

(d) Period, $T = T_1 + T_2 = R_x C_x(\ln 2 + \ln 3)$

$= R_x C_x \ln 6$

frequency $= 100Hz$

$\therefore \; R_x C_x \ln 6 = \frac{1}{100}$

$\Rightarrow C_x = \frac{1}{100 R_x \ln 6} = \frac{1}{100 \times 558 \times \ln 6}$

$\boxed{C_x = 10\,uF} = 1\times 10^{-5}\,F$

Set-B Question

(a) inverting Schmitt Trigger

(b) Higher threshold, $V_{TH} = \frac{R_1}{R_1+R_2} V_H$

$= \frac{10k}{10k+30k} \times 8V$

$\boxed{V_{TH} = 2V}$

Lower threshold, $V_{TL} = \frac{R_1}{R_1+R_2} V_L$

$= \frac{10k}{10k+30k} \times -16V$

$\boxed{V_{TL} = -4V}$

(c) High time $T_1 = R_x C_x \ln\left|\frac{V_H - V_{TL}}{V_H - V_{TH}}\right|$

$= R_x C_x \ln\left(\frac{8+4}{8-2}\right)$

$= R_x C_x \ln 2$

Low time, $T_2 = R_x C_x \ln\left|\frac{V_L - V_{TH}}{V_L - V_{TL}}\right|$

$= R_x C_x \ln\left(\frac{-16-2}{-16+4}\right)$

$= R_x C_x \ln 1.5$

Duty cycle, $D = \frac{T_1}{T_1+T_2} = \frac{\ln 2}{\ln 2 + \ln 1.5}$

$= 0.6309 = \boxed{63.09\%}$

(d) Period, $T = T_1 + T_2 = R_x C_x(\ln 2 + \ln 1.5)$

$= R_x C_x \ln 3$

frequency $= 1000Hz$

$\therefore \; R_x C_x \ln 3 = \frac{1}{1000}$

$\Rightarrow C_x = \frac{1}{1000 R_x \ln 3} = \frac{1}{1000 \times 910 \times \ln 3}$

$\boxed{C_x = 1\,uF} = 1\times 10^{-6}\,F$

Set (A)

Q2

(a) $V_{in} = \text{low} = -1.8\,V$

Q_1 OFF and Q_2 ON.

$V_{01} = \text{low} = -1.8\,V$

So, $V_B(Q_4) = -1.8 + 0.7 = -1.1\,V$

$$I_{Rc2} = \frac{0-(-1.1)}{0.75K} = 1.467\ mA = I_E$$

$\boxed{I_E = 1.467\ mA}$

(b) $V_{in} = \text{High}$

Q_1 ON, Q_2 OFF

$V_{02} = \text{low} = -1.8\,V$

$V_B(Q_3) = -1.1\,V$

$$I_{Rc1} = I_E = \frac{0-(-1.1)}{R_{c1}}$$

$$\Rightarrow R_{c1} = \frac{1.1\ V}{1.467\ mA} = \boxed{0.75\,K\Omega}$$

(c) $V_{in} = \text{High} = V_{01}$

Q_1 ON, Q_2 OFF

$V_B(Q_4) = 0\,V$

$V_{BE}(Q_4) = 0.7\,V$

$\Rightarrow V_E(Q_4) = V_{01} = -0.7\,V$

$$V_R = \frac{-1.8 - 0.7}{2} = -1.25\,V$$

(d) $V_{in} = \text{low} = -1.8\,V$

$V_{01} = -1.8\,V$, $V_{02} = -0.7\,V$

$$I_3 = \frac{(-0.7)-(-5.2)}{3k} = 1.5\ mA \quad ; \quad I_4 = \frac{(-1.8)-(-5.2)}{3K} = 1.13\,mA$$

$I_E = 1.467\ mA$.

$$P = \{0-(-5.2)\}\,[I_3 + I_4 + I_E] = 21.3044\ mW$$

Set (B)

(a) $V_{in} = \text{High}$

$V_{02} = \text{low} = -1.8\,V$

$V_B(Q_4) = -1.8 + 0.7 = -1.1\,V$

$$I_{Rc1} = \frac{0-(-1.1)}{0.7K\Omega} = \boxed{1.571\ mA = I_E}$$

(b) $V_{in} = \text{low} = V_{01} = -1.8\,V$

$V_B(Q_3) = -1.1\,V$

$$I_{Rc2} = I_E = \frac{0-(-1.1)}{R_{c2}} \Rightarrow R_{c2} = \frac{1.1\ V}{1.571\ mA} = 0.7\,K\Omega$$

(c) $V_{in} = \text{High} \Rightarrow V_{01} = \text{High}$

Q_1 ON, Q_2 off

$V_B(Q_3) = 0\,V$

As, $V_{BE} = 0.7\,V \Rightarrow V_{01} = -0.7\,V = \text{High}$

$$V_R = \frac{(-1.8)+(-0.7)}{2} = -1.25\,V$$

(d) $V_{in} = \text{HIGH} = -0.7\,V$

$V_{01} = -0.7\,V$, $V_{02} = -1.8\,V$

$$I_4 = \frac{-1.8-(-5.2)}{3K} = 1.13\ mA \quad ; \quad I_3 = \frac{(-0.7)-(-5.2)}{3K} = 1.5\ mA$$

$I_E = 1.571\ mA$.

$$P = \{0-(-5.2)\}\,\{I_4 + I_3 + I_E\} = 21.8452\ mW$$

Set-A

a)

$$V_O = -V_{REF}(b_2 + \frac{b_1}{2} + \frac{b_0}{4})$$

V_high = -4V

Encoding	VO
000	0
001	1
010	2
011	3
100	4
101	5
110	6
111	7

b)

Vref = 16V

V1 = 16/8R * R/2

V2 = 16/8R * 3R/2

Levels	Voltage	
V1	1	Range will be V2<Vin<V3
V2	3	
V3	5	
V4	7	
V5	9	
V6	11	
V7	13	

For DAC, input combinations will be 011, 100, 101

C)

4 Resistors
3 Comparators

Set-B

a)

$$V_O = -V_{REF}(b_2 + \frac{b_1}{2} + \frac{b_0}{4})$$

V_high = -8V

Encoding	VO
000	0
001	2
010	4
011	6
100	8
101	10
110	12
111	14

b)

Vref = 32V

V1 = 32/8R * R/2

V2 = 32/8R * 3R/2

Levels	Voltage	
V1	2	
V2	6	Range will be V3<Vin<V4
V3	10	
V4	14	
V5	18	
V6	22	
V7	26	

For DAC, input combinations will be 101, 110, 111

C)

4 Resistors
3 Comparators

Set A Question 4

(a) $\beta_F = 30$, $V_{CE(sat)} = 0.1\,V$, $\beta_R = 0.1$, $V_X = V_Y = V_Z = 6\,V$

$$I_1 = \frac{6 - 2.3}{R_1} = \frac{6-2.3}{10} = \boxed{0.37\ mA.}$$

$$I_X = I_Y = I_Z = \beta_R I_1 = 0.1 \times 0.37 = 0.037\ mA.$$

$$\therefore I_{B2} = I_1 + 3I_X = \boxed{0.481\ mA}$$

$$I_2 = \frac{V_{CC} - 0.9}{R_2} = \frac{6-0.9}{4} = \boxed{1.275\ mA.}$$

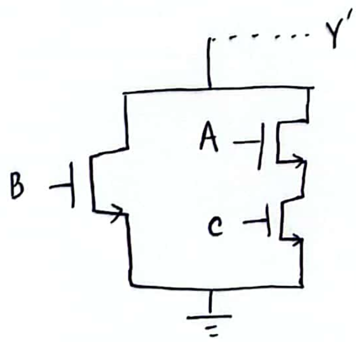
$$I_{E2} = I_2 + I_{B2} = 1.756\ mA.$$

$$I_B = \frac{0.8-0}{R_B} = \frac{0.8}{2} = \boxed{0.4\ mA}$$

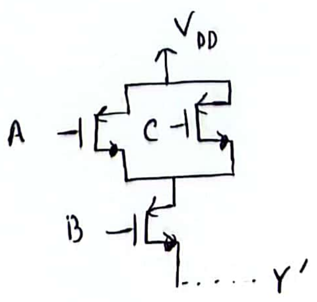
$$\therefore I_{B0} = I_{E2} - I_B = 1.756 - 0.4 = \boxed{1.356\ mA}$$

$$I_3 = \frac{6-0.1}{R_C} = \frac{6-0.1}{5} = \boxed{1.18\ mA}$$

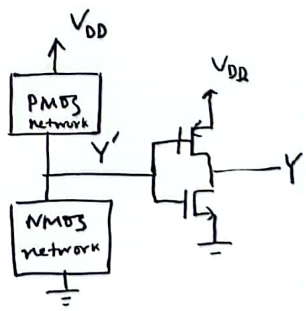
(b) $Y = AC + B$. Assume, $Y' = \overline{AC+B}$



NMOS network



PMOS network.



Set B Question 3

(a) $\beta_F = 30$, $V_{CE(sat)} = 0.1$, $\beta_R = 0.2$, $V_X = V_Y = V_Z = 6\,V$

$$I_1 = \frac{6-2.3}{R_1} = \boxed{0.37\ mA}$$

$$I_X = I_Y = I_Z = \beta_R I_1 = 0.2 \times 0.37 = 0.074\ mA.$$

$$\therefore I_{B2} = I_1 + 3I_X = \boxed{0.592\ mA}$$

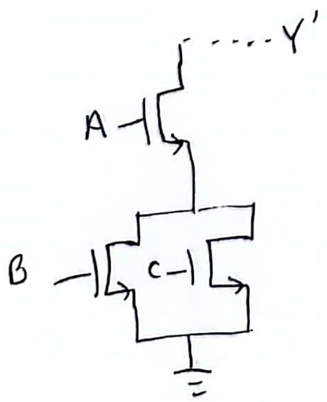
$$I_2 = \frac{V_{cc} - 0.9}{R_2} = \boxed{1.275\ mA}$$

$$I_{E2} = I_2 + I_{B2} = 1.867\ mA.$$

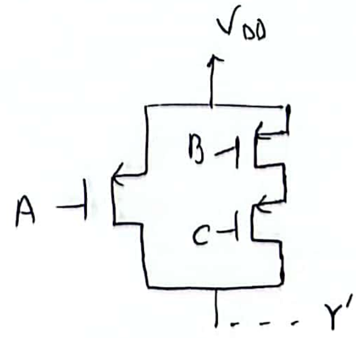
$$I_B = \frac{0.8}{2} = \boxed{0.4\ mA}, \quad I_{B0} = I_{E2} - I_B = \boxed{1.467\ mA}$$

$$I_3 = \frac{6-0.1}{R_C} = \frac{6-0.1}{6} = \boxed{0.983\ mA}$$

(b) $Y = A(B+C)$, Assume, $Y' = \overline{A(B+C)}$



NMOS network



PMOS network